

Supplemental material of “Statistical foundations of Rossmo’s formula with an application to the Yorkshire Ripper case”

The proofs below are written to be accessible to readers without a specialist background in mathematical statistics. Each proof follows the same structure: we first explain in plain language why the result should be true, then give the formal argument. Technical results from the statistical literature are cited rather than re-derived.

A. Proof of Theorem 1 (Existence and uniqueness of the population minimiser)

The energy $\mathcal{E}_{\alpha,\beta}^\infty$ has two terms. The first pulls the minimum toward the centre of the crime distribution; the second pushes it away from the buffer boundary. Under the generative model (2), both terms point to the same location: the true anchor point z^* . There is therefore no tension between them, and the minimum is unique.

Write $\mathcal{E}_{\alpha,\beta}^\infty(z) = \Phi(z) + \Psi(z)$, where

$$\Phi(z) = \frac{1}{\alpha} \mathbb{E} \|X - z\|$$

and

$$\Psi(z) = \beta \mathbb{E} [(B - \|X - z\|)_+^2].$$

Step 1: Φ has a unique minimum at z^* . The functional $\Phi(z) = \alpha^{-1} \mathbb{E} \|X - z\|$ is the classical *spatial median* or *Fermat–Weber functional* (Weiszfeld 1937; Kemperman 1987). It is strictly convex whenever the distribution of X is not concentrated on a single straight line; this is a non-trivial consequence of the geometry of the Euclidean norm and is established in Milasevic and Ducharme (1987), from which the uniqueness result also follows: see Weiszfeld (1937) for the classical (non-strict) convexity argument. Under model (2), the crimes are spread over the annulus $\{\|x - z^*\| > B\}$, which is not contained in any line, so Φ is strictly convex with a unique minimum. To see that the minimum is at z^* , note that $f_{z^*,\alpha,B}$ is radially symmetric around z^* : for every direction, there is equal mass on both sides. The unique minimiser of Φ is z^* : since $f_{z^*,\alpha,B}$ is radially symmetric about z^* , the gradient $\nabla_z \Phi(z^*) = -\alpha^{-1} \mathbb{E}[(X - z^*)/\|X - z^*\|]$ equals zero by isotropy (the unit directions cancel), so z^* is a critical point; combined with the strict convexity established above, it is the unique minimiser. See Milasevic and Ducharme (1987) for the general uniqueness result.

The energy $\mathcal{E}_{\alpha,\beta}^\infty$ is Lipschitz-continuous on Ω with constant $L = \alpha^{-1} + 2\beta B$: for any $z, z' \in \Omega$,

$$\begin{aligned} |\ell_{\alpha,\beta}(z, x) - \ell_{\alpha,\beta}(z', x)| &\leq \frac{\|z - z'\|}{\alpha} + 2\beta B \|z - z'\| \\ &= L \|z - z'\|, \end{aligned}$$

using the triangle inequality for the distance term and the Lipschitz property of $(t)_+^2$ near the buffer. Taking expectations preserves the bound.

Step 2: Ψ is zero at z^* and non-negative elsewhere. Under model (2), every crime satisfies $\|X - z^*\| > B$ almost surely, so $(B - \|X - z^*\|)_+ = 0$ almost surely, giving $\Psi(z^*) = 0$. Since $(t)_+^2 \geq 0$ for all t , we have $\Psi(z) \geq 0$ for every z .

Step 3: Combining. For any $z \neq z^*$: $\Phi(z) > \Phi(z^*)$ by strict convexity, and $\Psi(z) \geq 0 = \Psi(z^*)$. Therefore $\mathcal{E}_{\alpha,\beta}^\infty(z) > \mathcal{E}_{\alpha,\beta}^\infty(z^*)$, confirming that z^* is the unique minimiser. Existence follows from continuity of $\mathcal{E}_{\alpha,\beta}^\infty$ on the compact set Ω (Weierstrass’s theorem: see Rudin (1976, Theorem 4.16)). $\square$

B. Proof of Theorem 2 (Γ -convergence to Rossmo’s formula)

As β grows, the soft buffer penalty $(B - d)_+^2$ becomes increasingly severe. In the limit $\beta \rightarrow \infty$, any candidate location that falls within the buffer of a crime site incurs infinite cost and is therefore excluded. This is exactly Rossmo’s hard-buffer rule: the hit score is zero inside the buffer and positive outside. Γ -convergence is the mathematical tool that formalises this intuition and guarantees that the minimisers also converge (Braides 2002; Dal Maso 1993).

We verify the two conditions of Γ -convergence (Braides 2002).

Let $z_k \rightarrow z$ and $\beta_k \uparrow \infty$. Fix a crime site x . If $\|x - z\| > B$ (outside the buffer), then for large k , $\|x - z_k\| > B$ by continuity, so the buffer penalty is zero and $\ell_{\alpha,\beta_k}(z_k, x) = \|x - z_k\|/\alpha \rightarrow \|x - z\|/\alpha = \ell_{\alpha,\infty}(z, x)$. If $\|x - z\| < B$ (inside the buffer), then for large k , $\|x - z_k\| < B - \delta$ for some $\delta > 0$, giving $\ell_{\alpha,\beta_k}(z_k, x) \geq \beta_k \delta^2 \rightarrow \infty = \ell_{\alpha,\infty}(z, x)$. Applying Fatou’s lemma (Ash and Doléans-Dade 2000, Theorem 1.6.8) to pass from pointwise to expected values gives the liminf condition.

At $z = z^*$, the support of $f_{z^*,\alpha,B}$ lies entirely outside the buffer, so $\ell_{\alpha,\beta_k}(z^*, X) = \|X - z^*\|/\alpha$ for all k , and the expected energy equals $\mathcal{E}_{\alpha,\infty}^\infty(z^*)$ for every β_k . Taking the constant sequence $z_k \equiv z^*$ gives the required recovery sequence.

To complete the Γ -convergence argument, we supply recovery sequences for all $z \in \Omega$:

Case 1: $\|x - z\| > B$ for all x in the support. Take $z_k \equiv z$; then $\ell_{\alpha,\beta_k}(z, x) = \|x - z\|/\alpha = \ell_{\alpha,\infty}(z, x)$ for all k , so the limsup condition holds with equality.

Case 2: $\|x - z\| < B$ for some x . Take $z_k \equiv z$; then $\ell_{\alpha,\beta_k}(z, x) \geq \beta_k (B - \|x - z\|)^2 \rightarrow +\infty = \ell_{\alpha,\infty}(z, x)$.

Case 3: $\|x - z\| = B$ for some x . Take $z_k \equiv z$; then $\ell_{\alpha,\beta_k}(z, x) = \|x - z\|/\alpha = B/\alpha = \ell_{\alpha,\infty}(z, x)$ for all k . In all three cases the constant sequence serves as a valid recovery sequence.

Convergence of minimisers follows from the fundamental theorem of Γ -convergence (Braides 2002, Theorem 1.21), which applies because Ω is compact (equicoercivity is automatic). $\square$

C. Proof of Theorem 3 (Consistency and asymptotic normality)

By the law of large numbers, the empirical energy $\mathcal{E}_{\alpha,\beta}^n(z)$ converges to the population energy $\mathcal{E}_{\alpha,\beta}^\infty(z)$ as $n \rightarrow \infty$, uniformly over all candidate locations z . Since $\mathcal{E}_{\alpha,\beta}^\infty$ has a unique minimum at z^* (Theorem 1), the minimiser of the empirical energy must also approach z^* . The rate $1/\sqrt{n}$ and the Gaussian shape of the error follow from the central limit theorem applied to the gradient of the energy.

Consistency. We apply the general consistency theorem for M -estimators (Newey and McFadden 1994, Theorem 2.1), which requires four conditions: (M1) Ω is compact (given by Assumption H1); (M2) $\mathcal{E}_{\alpha,\beta}^\infty$ has a unique minimiser (Theorem 1); (M3) the loss $\ell_{\alpha,\beta}(z, x)$ is continuous in z and satisfies $\mathbb{E}[\sup_z |\ell_{\alpha,\beta}(z, X)|] < \infty$ (verified using the Lipschitz bound from Step 1 of Theorem 1 and the finite mean of $\|X\|$ under model (2)); (M4) the empirical energy converges uniformly to the population energy, i.e., $\sup_{z \in \Omega} |\mathcal{E}_{\alpha,\beta}^n(z) - \mathcal{E}_{\alpha,\beta}^\infty(z)| \rightarrow 0$ in probability.

Condition (M4) follows from a uniform law of large numbers. The loss class $\{\ell_{\alpha,\beta}(z, \cdot) : z \in \Omega\}$ is Lipschitz in z with constant $L = \alpha^{-1} + 2\beta B$, so its metric entropy is of order $O(\varepsilon^{-2})$, satisfying the Glivenko–Cantelli condition for Lipschitz classes (van der Vaart and Wellner 1996, Theorem 2.4.1). All four conditions hold, so $\hat{z}_n \xrightarrow{\mathbb{P}} z^*$.

Asymptotic normality. We apply van der Vaart (1998, Theorem 5.23). At $z = z^*$, the buffer penalty vanishes almost surely (Step 2 of Theorem 1), so locally $\mathcal{E}_{\alpha,\beta}^\infty(z) \approx \alpha^{-1} \mathbb{E} \|X - z\|$. Differentiating twice under the expectation (Folland 1999, Theorem 2.27):

$$H = \nabla_z^2 \mathcal{E}_{\alpha,\beta}^\infty(z^*) \\ = \frac{1}{\alpha} \mathbb{E} \left[\frac{1}{\|X - z^*\|} \left(I_2 - \frac{(X - z^*)(X - z^*)^\top}{\|X - z^*\|^2} \right) \right].$$

Because $f_{z^*,\alpha,B}$ is radially symmetric, the unit direction $u = (X - z^*)/\|X - z^*\|$ is uniformly distributed on the unit circle, giving $\mathbb{E}[uu^\top] = \frac{1}{2} I_2$ (Mardia and Jupp 2000). Computing $\mathbb{E}[\|X - z^*\|^{-1}]$ by integration in polar coordinates against $f_{z^*,\alpha,B}$ yields $(\alpha + B)^{-1}$, so

$$H = \frac{1}{2\alpha(\alpha + B)} I_2, \quad \Sigma = \mathbb{E}[\nabla_z \ell (\nabla_z \ell)^\top] = \frac{1}{2\alpha^2} I_2,$$

and the sandwich variance is $V_{\alpha,\infty} = H^{-1} \Sigma H^{-1} = 2(\alpha + B)^2 I_2$. By van der Vaart (1998, Theorem 5.23), $\sqrt{n}(\hat{z}_n - z^*) \xrightarrow{d} \mathcal{N}(0, V_{\alpha,\infty})$. $\square$

D. Proof of Theorem 4 (Degenerate-buffer theorem)

Rossmo’s hard buffer creates a sharp boundary at distance B from each crime site: candidate locations just outside the buffer receive a finite score, while those just inside receive zero. This discontinuity means that a small shift in the crime sites—caused, for instance, by recording imprecision or by the discovery of a new crime—can abruptly change which locations are admissible, destabilising the maximum. When two crime sites are close enough that their buffers overlap, the admissible region develops a complex shape with no guarantee of a unique maximum.

Part (i): discontinuity of the loss. Fix x_0 and let z_0 satisfy $\|x_0 - z_0\| = B$. Then $\ell_{\alpha,\infty}(z_0, x_0) = B/\alpha < \infty$. But any sequence $z_k \rightarrow z_0$ with $\|x_0 - z_k\| < B$ gives $\ell_{\alpha,\infty}(z_k, x_0) = +\infty$. Hence $z \mapsto \ell_{\alpha,\infty}(z, x_0)$ is discontinuous at z_0 , violating the continuity condition (M3) of the general consistency theorem for M -estimators (Newey and McFadden 1994, Theorem 2.1).

Part (ii): non-unique maximiser. We study the two cases.

Case (a): single crime site. For a single crime site x , Rossmo’s score function $z \mapsto \psi(\|z - x\|)$ depends on z

only through $d = \|z - x\|$. For $d > B$ the score d^{-f} is strictly decreasing in d ; for $d < B$ the score $B^{g-f}(2B - d)^{-g}$ is strictly increasing in d (since $2B - d$ decreases as d increases toward B). Both arms therefore achieve their maximum as $d \rightarrow B$, and both equal B^{-f} at $d = B$ (by the continuity factor B^{g-f}). Hence $\psi(d)$ is maximised at exactly $d = B$, and the set of maximisers is $\{z \in \Omega : \|z - x\| = B\}$ —the full circle of radius B centred at x .

Case (b): two crime sites with overlapping buffers. Place two crime sites at $x_1 = (-r, 0)$ and $x_2 = (r, 0)$ with $0 < r < B$, so the buffers $B_B(x_1)$ and $B_B(x_2)$ overlap. The admissible region $\mathcal{Z} = \Omega \setminus (B_B(x_1) \cup B_B(x_2))$ consists of two symmetric “lune” regions left and right of the overlapping discs. On the y -axis, the symmetry of the configuration gives $d(z, x_1) = d(z, x_2) = \sqrt{r^2 + t^2}$ for $z = (0, t)$. A point $(0, t)$ belongs to $\mathcal{Z}$ if and only if $\sqrt{r^2 + t^2} > B$, i.e. $|t| > \sqrt{B^2 - r^2}$. Rossmo’s score restricted to the y -axis is

$$p_R(0, t \mid x_1, x_2) = \frac{2k}{(r^2 + t^2)^{f/2}},$$

which is strictly decreasing in $|t|$ (since $f > 0$). Therefore this restricted score is maximised at the smallest admissible $|t|$, namely $|t| = \sqrt{B^2 - r^2}$, giving the two distinct maximisers $z_\pm = (0, \pm\sqrt{B^2 - r^2})$ with identical scores. These points are also global maximisers over all of $\mathcal{Z}$. For any z in the closure of $\mathcal{Z}$, both distances satisfy $d(z, x_1) \geq B$ and $d(z, x_2) \geq B$ (by definition of the admissible region). Since $t \mapsto t^{-f}$ is strictly decreasing on $(0, \infty)$, we have $d_i^{-f} \leq B^{-f}$ for each i , with equality if, and only, if $d_i = B$. Therefore

$$p_R(z \mid x_1, x_2) = k \left(\frac{1}{d_1^f} + \frac{1}{d_2^f} \right) \leq 2k B^{-f},$$

with equality if and only if $d_1 = d_2 = B$ simultaneously. The system $\|z - x_1\| = B$, $\|z - x_2\| = B$ is satisfied exactly at $z_\pm = (0, \pm\sqrt{B^2 - r^2})$. Hence $\arg \max_z p_R(z \mid x_1, x_2) = \{z_+, z_-\}$, a set containing two points. $\square$

E. Proof of Theorem 5 (Properties of the sigmoid-smooth score)

The sigmoid function $\sigma_\tau(t) = 1/(1 + e^{-t/\tau})$ is infinitely differentiable and interpolates smoothly between 0 and 1. When it replaces the hard indicator in Rossmo’s formula, it inherits that smoothness everywhere. The key observation for Part (ii) is that Rossmo’s formula was already designed so that the two arms of the score agree in value at $d = B$ —that is precisely the role of the factor B^{g-f} . Because the two arms agree in value at the junction point, their smooth weighted average by σ_τ is also smooth there, and the derivative jump disappears.

Part (i): Smoothness. The logistic function σ_τ is C^∞ on $\mathbb{R}$ (Rudin 1976, Chapter 8). The distance $d_i = \|z - x_i\|$ is C^∞ on $\Omega \setminus \{x_i\}$. On the domain $\Omega \setminus \{z : d(z, x_i) \in \{0, 2B\}\}$, the score $S_\tau(z)$ is a finite sum of products and compositions of C^∞ functions, hence C^∞ there (Rudin 1976, Theorem 9.15). The restriction to this domain excludes a set of measure zero in Ω and does not affect the practical use of gradient-based optimisation.

Part (ii): No slope jump at $d = B$. Denote the two arms as $\phi_{\text{far}}(d) = (d + \varepsilon)^{-f}$ and $\phi_{\text{near}}(d) = B^{g-f}((2B -$

$d)_+ + \varepsilon)^{-g}$, so that each crime's contribution is

$$\psi_\tau(d) = \sigma_\tau(d - B) \phi_{\text{far}}(d) + (1 - \sigma_\tau(d - B)) \phi_{\text{near}}(d).$$

Differentiating:

$$\begin{aligned} \psi'_\tau(d) &= \dot{\sigma}_\tau(d - B) [\phi_{\text{far}}(d) - \phi_{\text{near}}(d)] \\ &\quad + \sigma_\tau(d - B) \phi'_{\text{far}}(d) + (1 - \sigma_\tau(d - B)) \phi'_{\text{near}}(d), \end{aligned}$$

where $\dot{\sigma}_\tau(d - B) = d/dd[\sigma_\tau(d - B)] = \sigma_\tau(d - B)(1 - \sigma_\tau(d - B))/\tau > 0$. At $d = B$: the factor B^{g-f} in Rossmo's formula ensures $\phi_{\text{far}}(B) = \phi_{\text{near}}(B)$, so the first term vanishes exactly. Since $\sigma_\tau(0) = 1/2$, the remaining terms give

$$\psi'_\tau(B) = \frac{\phi'_{\text{far}}(B) + \phi'_{\text{near}}(B)}{2},$$

the average of the two one-sided derivatives of Rossmo's formula—a finite, well-defined value, confirming differentiability at $d = B$ for every $\tau > 0$.

Part (iii): Γ -convergence to Rossmo. For fixed z with $d_i \neq B$ for all i , the logistic function satisfies

$$\sigma_\tau(d_i - B) = \frac{1}{1 + e^{-(d_i - B)/\tau}} \longrightarrow \mathbf{1}_{\{d_i > B\}}$$

pointwise as $\tau \rightarrow 0^+$ (a standard property of the logistic function: for fixed $d_i \neq B$, $e^{-(d_i - B)/\tau} \rightarrow 0$ if $d_i > B$ and $e^{-(d_i - B)/\tau} \rightarrow +\infty$ if $d_i < B$). Since $S_\tau(z)$ is a *finite* sum of n terms and each term converges pointwise (as shown above), $S_\tau(z) \rightarrow p_R(z)$ follows by the continuity of addition. The liminf and recovery conditions of Γ -convergence (Braides 2002) follow from this pointwise convergence and the uniform bound $|S_\tau(z)| \leq \sum_i \max(\phi_{\text{far}}(d_i), \phi_{\text{near}}(d_i)) < \infty$ on the compact set Ω . Points where $d_i = B$ for some i form a set of measure zero in Ω and do not affect the Γ -limit. $\square$